🧠 AI-Based Driver Monitoring System (DMS) – Use Case for Explainability

Quick Introduction:
This use case explores explainability requirements for AI models used in Driver Monitoring Systems (DMS), which are mandated by the **EU General Safety Regulation (EU) 2019/2144**. Daimler Truck intends to leverage its DMS with an AI model to reach a 5-Star Rating within NCAP's category Driver Engagement. The AI model shall ensure to assess, in an error-free manner, driver attention, fatigue, and distraction levels using in-cabin cameras and sensor data. The goal is to ensure that the system's decisions are transparent, trustworthy, and auditable.

The AI-enhanced DMS is a safety-critical system that must balance **real-time responsiveness** with **interpretable behavior**, especially when issuing warnings or initiating interventions.

Regulatory Context:

The DMS is required under the **EU General Safety Regulation (EU) 2019/2144**, which mandates the following systems:

- **Driver Drowsiness and Attention Warning (DDAW):**
  Required for all new type-approvals from **July 2022** and all vehicles produced (new type-approval might required) from **July 2024**. It assesses driver alertness through vehicle behavior and issues warnings when drowsiness is detected
- **Advanced Driver Distraction Warning (ADDW):**
  Required for new type-approvals from **July 2026** and all vehicles produced (see above) from **July 2028**. It uses **direct driver monitoring** (e.g., eye tracking) to detect visual distraction and warn the driver accordingly

These systems must meet specific performance, HMI, and validation requirements as outlined in the delegated regulations

System Overview:

The AI-powered DMS consists of three main components, two of which are AI-driven:

- **Perception Module (AI-driven)**: Uses computer vision models (e.g., CNNs or transformer-based architectures) to extract facial landmarks, eye gaze, head pose, and eyelid closure from infrared or RGB camera input. These features are processed at the pixel and frame level to generate structured behavioral indicators.
- **Cognitive State Estimation Module (AI-driven)**: Applies temporal AI models (e.g., LSTMs or attention-based networks) to sequences of extracted features to infer driver

states such as drowsiness, distraction, or cognitive overload. These models are trained on annotated datasets and output probabilistic classifications.

- **Decision and Alerting Module (rule-based)**: This module is not AI-based. It uses predefined logic and thresholds to determine the appropriate system response—ranging from visual/auditory alerts to escalation protocols—based on the AI-inferred driver state and contextual driving parameters.

AI Requirements (for participant elaboration):

- Req. 1: The AI model shall detect and classify driver states (e.g., attentive, distracted, drowsy) using observable facial and behavioral cues (e.g., gaze direction, eyelid closure) extracted from camera input.
- Req. 2: The AI model shall operate reliably across diverse lighting conditions, facial features, and driver behaviors by applying robust preprocessing and normalization techniques.
- Req. 3: The AI model shall integrate data from multiple sources (e.g., camera, vehicle dynamics) at the feature level to improve detection accuracy through sensor fusion.
- Req. 4: The AI model shall provide interpretable outputs (e.g., gaze direction, eye closure duration, confidence scores) that can be used by the rule-based system to generate real-time visual and auditory alerts.
- Req. 5: The AI model shall log detection events, contributing features, and model confidence levels for post-event analysis and regulatory compliance.
- Req. 6: The AI model shall minimize false alerts and missed detections by applying calibrated confidence thresholds and validated decision boundaries.

Example Scenario:

A driver looks away from the road for 4 seconds while approaching a pedestrian crossing.

- The DMS detects prolonged gaze deviation and triggers a yellow visual alert: *"Please focus on the road – eyes off road for 4.0s."*
- If the behavior continues, a red alert with sound is issued.
- The system logs the event with a timestamp, detected features, and model confidence for later review.